SOFTWARE ENGINEER INTERVIEW

Software Engineering & System Simulation

Production Software Development · GPU LLM Serving Research

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Education & Professional Experience

Education

National Taiwan University
M.S. Candidate, Data Science Master's Degree Program
Expected Graduation: June 2027

National Chengchi University
B.S. in Computer Science
Double Major in Electronic Physics
Sep. 2021 – Jun. 2025

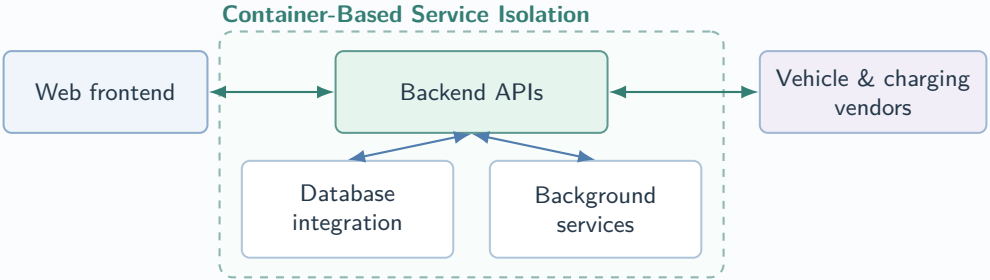
Professional Experience

AIMobile Co., Ltd.
Inventec Group
Contract Software Engineer
Apr. 2025 – Jul. 2025

Neopower Technologies Co., Ltd.
Contract Software Engineer
Jun. 2023 – Mar. 2025

NeoPower: Backend System Architecture

Electric Bus Charging Management Platform



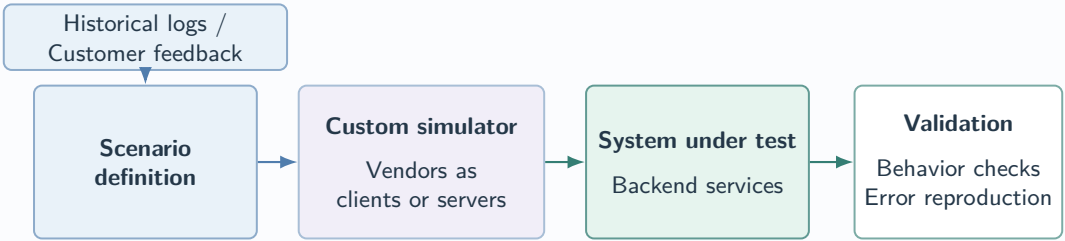
My contributions

System planning & backend development / Database integration

Log analysis & debugging / Deployment & maintenance

NeoPower: Scenario-Based System Simulation

I developed a custom simulator for vendor interaction and failure reproduction.



High Load

Many buses arrive at once

Network Failure

Mass disconnection of stations / chargers

NeoPower: Testing & Production Reliability

Pre-Deployment Testing

Simulator-based validation

Normal / abnormal conditions
Regression testing

Tests pass

Deploy updated system

Post-Deployment Validation

Charging-power allocation algorithms

Cross-Site Comparison: similar operating conditions

Some sites
Previous algorithm

Other sites
Updated algorithm

Compare operating behavior

System logs / operating data
Customer feedback

LLM Serving: Heterogeneous GPU & Memory

How can request scheduling balance performance and energy across heterogeneous GPUs?

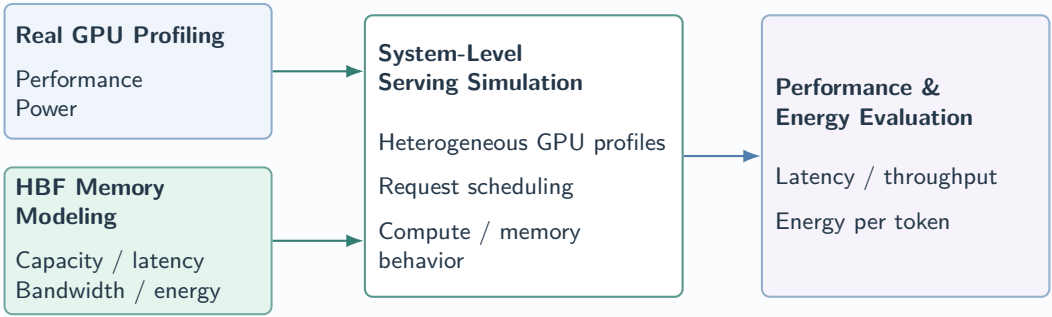
Prefill	**Decode**	**Model weights**	**KV cache**
compute-heavy	memory-sensitive	large, read-mostly	latency-sensitive

Conceptual compute / memory profiles

	HBM-only fast memory	**HBF-dominant** high capacity	**Hybrid**
High compute	Latency-first memory	Capacity-first memory	HBM + HBF split by role
Low compute	Latency-first memory	Capacity-first memory	HBM + HBF split by role

Research Methodology: Simulation Framework

Proposed framework: evaluate serving performance and energy.



Current Progress & Research Roadmap

Completed

Literature review

Research problem & design-space exploration

Evaluation of candidate simulation frameworks

Planned

Reproduce serving baselines

Profile real GPU performance & power

Integrate HBF models

Explore energy-aware scheduling
and evaluate trade-offs

Thank You

Questions & Discussion